



Optimization Theory and Methods

2026 Autumn  同济经管
TONGJI SEM

魏可伧

kejiwei@tongji.edu.cn

<https://kejiwei.github.io/>


中国高质量MBA教育认证

 AACSB
ACCREDITED

 EQUIS
ACCREDITED

- Mixed-Integer Optimization for *Portfolio Management*.
- Two-Stage Game Theory Model for *Scheduling Coast Guard Patrol*.
- Queuing and Simulation Models for *Energy Audit Program*.
- Queuing and Simulation Models for *Car Queues* through a Security Gate.
- Discrete Event Simulation to Reduce *Healthcare-Associated Infections*.
- Network Optimization based Auction for *Improving School Meals in Chile*.
- Solving Railroad Blocking Problem at Four Major US *Railroads*.
- Queuing Networks for *Improving Capacity and Lead Times* at IBM.
- *Modeling Overview*.

↳ Mixed-Integer Optimization for Portfolio Management

- **Mixed-Integer Optimization for Portfolio Management**, Grantham, Mayo, Van Otterloo and Company LLC (1999).
 - Investment management firm with \$26 billion on assets.
 - **Aim:** Create a portfolio that resembles a target portfolio, and
 - ✓ Has small number of assets and small number of transactions.
 - ✓ Has high returns, high liquidity and low transaction costs.
 - Modelled as a mixed-integer quadratic optimization problem.
 - Solved with a FORTRAN model using CPLEX optimizer.
 - Helped the firm keep existing clients' business and also launch two new funds successfully (\$1.1 billion in a year).
 - Reduced annual transaction costs by \$4 million.

↳ Two-Stage Game Theory Model for Scheduling Coast Guard Patrol

- **Two-Stage Game Theory Model for Scheduling Coast Guard Patrol ,**
United States Coast Guard (USCG 2013).
- Challenges from potential terrorists within the maritime environment:
Intent to blow up US oil tankers.
- Two-Stage Model using Attacker-Defender Stackelberg Game Framework.
- Defender problem approximated using a mixed-integer linear optimization formulation.
- Java + CPLEX implementation.
- Validated through human-subject experiments and simulations.
- Already deployed in the Port of Boston, and considered a success by USCG;
Efforts underway to deploy in New York.

↳ Queuing and Simulation Models for Energy Audit Program

- **Queuing and Simulation Models for Energy Audit Program, Florida Power and Light Company(1981).**
 - Offers for energy audits at residential customers' homes.
 - Aim is to determine the right number of mails offering customer audits and the right level of auditor staffing to serve the anticipated demand.
 - Multi-server queuing model and a discrete event simulation model programmed in FORTRAN.
 - Performance evaluated in terms of 1) average wait, 2) maximum wait, 3) backlog size, and 4) auditor utilization.
 - Model was successfully deployed and company used model recommendations to decide staffing and mailing levels.

↳ Queuing and Simulation Models for Car Queues through a Security Gate

- **Queuing and Simulation Models for Car Queues through a Security Gate, Westinghouse Hanford Company(1989).**
 - An average of 7 buses and 285 private vehicles entered the gate each morning. Average 2 guards on duty.
 - Queue overflow onto adjacent highway causing major traffic safety issues.
 - Alternatives were more guards or more traffic channels.
 - A discrete event simulation using queuing formulas built using manually collected data.
 - Recommendation of utilizing parallel channels was made.
 - Success during a brief trial period ensured that the change became permanent.

↳ Discrete Event Simulation to Reduce Healthcare-Associated Infections

■ Discrete Event Simulation to Reduce Healthcare-Associated Infections,

Cook County Hospital, Chicago IL (2009).

- Healthcare-Associated Infections (HAIs) responsible for 2 million infections and 100,000 deaths annually in the US.
- Discrete event simulation used to model the interaction between pathogens, patients and visitors in ICU.
- Models of process by which they enter an ICU, interact with health-care workers and with each other, infect, become infected, are cured, and are discharged.
- Both hand-hygiene and isolation of the infected reduce HAI rates, reduce capacity and increase costs. And they interact!
- Hand hygiene found to be more cost effective than isolation.

↳ **Network Optimization based Auction for Improving School Meals in Chile**

■ **Network Optimization based Auction for Improving School Meals in Chile,**

The Government of Chile(2002).

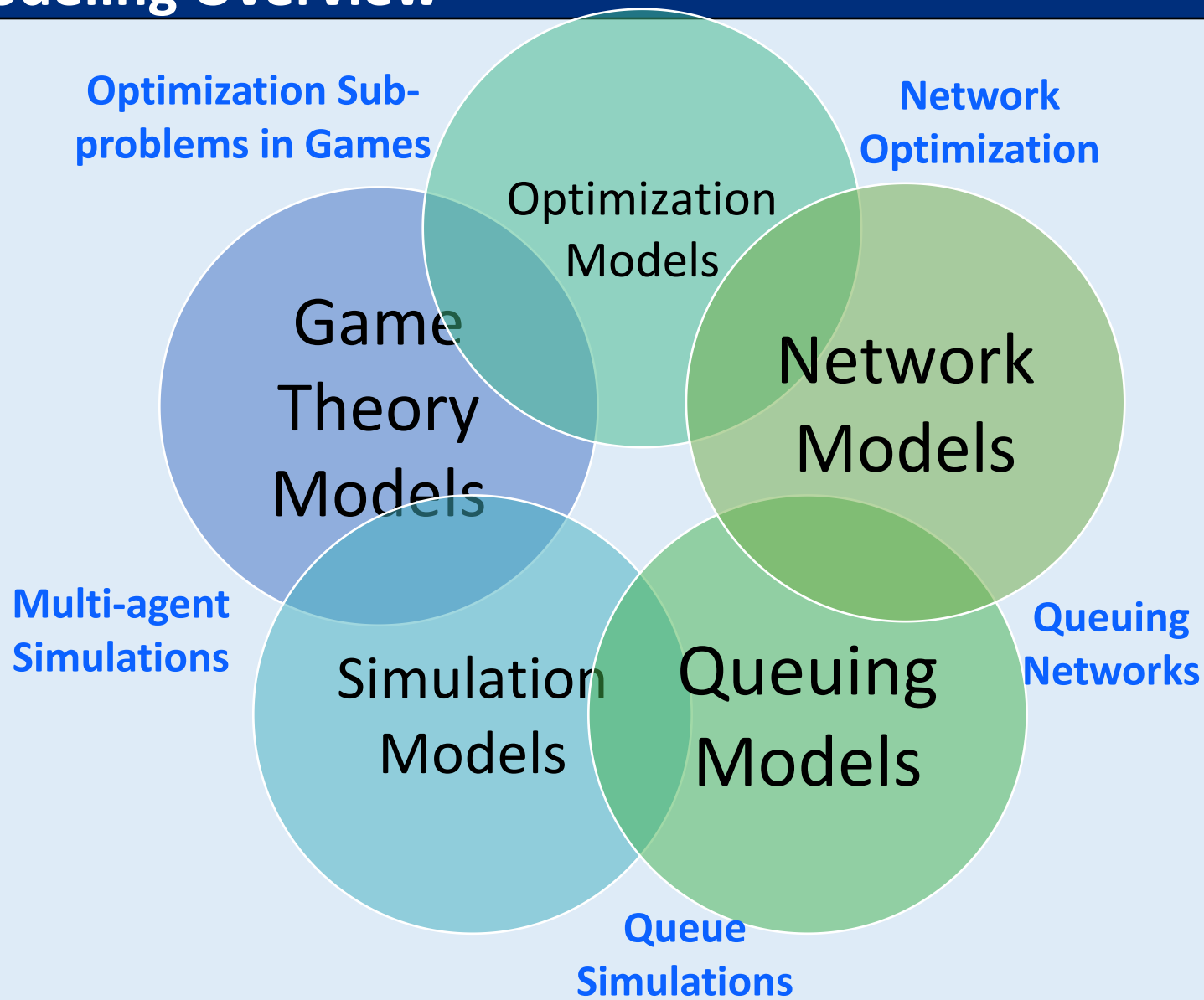
- Spends \$180 million a year to feed 1.3 million students from low-income families.
- Manual process of awarding meal contracts was inefficient.
- Designed a single-round sealed-bid combinatorial auction.
- Developed a network-based integer linear optimization model with >10,000 binary variables. Found optimal solution.
- Improved price-to-quality ratio of meals. Saved \$40 million annually (equivalent to cost of feeding 300,000 children).
- Successful test in 1997. Completely automated from 1999 onward.

↳ Solving Railroad Blocking Problem at Four Major US Railroads

- **Solving Railroad Blocking Problem at Four Major US Railroads**, three major freight companies in US: CSX Transportation, Norfolk Southern Corporation, and Burlington Northern Santa Fe Railway(2007).
 - A major challenge is to classify shipments into blocks, at railroad yards, to be transported together over the network.
 - Developed a multicommodity-flow, network-design and routing problem.
 - Solved using very large-scale neighborhood search heuristics.
 - Implemented at all three railway companies now.
 - Projected to save several hundred million dollars annually.

- **Queuing Networks for Improving Capacity and Lead Times at IBM**, IBM's semiconductor manufacturing factories in New York(2010).
 - Silicon wafers proceed through 500 to 1000 fabrication, inspection and test steps using multimillion dollar machines.
 - Two objectives are high utilization of machines to reduce costs and shorter lead times to expedite time to market.
 - Developed and implemented Enterprise Production Planning and Optimization System (EPOS): an advanced queuing network model for capacity planning.
 - Improved factory output by 8%. Saved >1500 sq. ft. of factory floor space. Saved >\$700,000 operating expenses per year. Reduced future capital needs by >\$30 million.

↳ Modeling Overview



Objective :

Key Concepts :